1. Inequalities (3.6, 4.7)

These bound a probability or an expectation by something you can compute. Use them when the exact distribution is unavailable.

Markov. If $X \ge 0$ and $r > 0$, then $P(X \ge r) \le EX/r$. Proof: $EX \ge E[X\,1_{\{X \ge r\}}] \ge r\,P(X \ge r)$.

Chebyshev. $P(|X-\mu| \ge r) \le \sigma^2/r^2$. Apply Markov to $(X-\mu)^2$. For an iid sample, $\mathrm{Var}(\bar X) = \sigma^2/n$, so $P(|\bar X - \mu| \ge \epsilon) \le \sigma^2/(n\epsilon^2)$.

Relative error (HW2 coin). X_i Bernoulli(p), $p \ge a \in (0,1)$, $\tilde p = \bar X$. Want $P(|\tilde p - p| \ge \epsilon p) \le \delta$. Chebyshev:

$$P \le \frac{p(1-p)}{n\epsilon^2 p^2} = \frac{1-p}{n\epsilon^2 p}.$$

$(1-p)/p$ decreases in p, so its worst value on $p \ge a$ is at $p = a$. Thus the bound is $(1-a)/(na\epsilon^2)$. Any

$$n \ge \frac{1-a}{a\epsilon^2\delta}$$

works. The information $p \ge a$ replaces the cruder $p(1-p) \le 1/4$.

Jensen (functional). If ϕ is convex, $\phi(EX) \le E\phi(X)$. If ϕ is strictly convex, equality holds iff X is constant a.s. Concave ϕ reverses the inequality. Proof: convexity puts a line under the graph, $\phi(x) \ge \phi(\mu) + a(x-\mu)$, with $\mu = EX$ and $a = \phi'(\mu)$ when ϕ is differentiable. Substitute X and take expectations. The term $aE(X-\mu)$ is 0, so $E\phi(X) \ge \phi(EX)$. Uses: $\mathrm{Var}\,X = EX^2 - (EX)^2 \ge 0$ because $t \mapsto t^2$ is convex; $E\log X \le \log EX$ because log is concave.

AM–GM from Jensen. For $x_i > 0$, log concave gives $\log\big((x_1+\cdots+x_n)/n\big) \ge (\log x_1 + \cdots + \log x_n)/n = \log(x_1\cdots x_n)^{1/n}$. Exponentiate: the arithmetic mean is at least the geometric mean, with equality iff all x_i agree.

Moments get stronger. For $0 < s < r$, $(E|X|^s)^{1/s} \le (E|X|^r)^{1/r}$, so $E|X|^r < \infty$ implies $E|X|^s < \infty$. Proof: $r/s > 1$, so $y \mapsto y^{r/s}$ is convex on $[0,\infty)$. Jensen on $Y = |X|^s$ gives $(E|X|^s)^{r/s} \le E|X|^r$. Raise both sides to s/r, then to $1/s$.

Cauchy–Schwarz. $E|XY| \le \sqrt{EX^2\,EY^2}$. Proof: for every real t, $0 \le E(|X| - t|Y|)^2 = EX^2 - 2t\,E|XY| + t^2EY^2$. If $EY^2 > 0$, choose $t = E|XY|/EY^2$. The quadratic in t is nonnegative for every t, so its discriminant is ≤ 0: $(E|XY|)^2 \le EX^2\,EY^2$. Apply the same inequality to $X - \mu_X$ and $Y - \mu_Y$ to get $|\mathrm{Cov}(X,Y)| \le \sigma_X\sigma_Y$, hence $\rho \in [-1,1]$. Equality iff $Y = a + bX$ a.s.

Hölder. For $p, q > 1$ with $1/p + 1/q = 1$, $E|XY| \le \|X\|_p \|Y\|_q$, where $\|X\|_p = (E|X|^p)^{1/p}$. Proof: Young's inequality from the concavity of log, $ab \le a^p/p + b^q/q$ for $a, b \ge 0$. Normalize $U = |X|/\|X\|_p$ and $V = |Y|/\|Y\|_q$ (the claim is trivial if either norm is 0). Then $E|UV| \le E(U^p/p + V^q/q) = 1/p + 1/q = 1$, which is $E|XY| \le \|X\|_p\|Y\|_q$. Cauchy–Schwarz is the case $p = q = 2$.

Minkowski. For $p \ge 1$, $\|X+Y\|_p \le \|X\|_p + \|Y\|_p$. Proof: $|X+Y|^p \le |X|\,|X+Y|^{p-1} + |Y|\,|X+Y|^{p-1}$. Take expectations and apply Hölder to each product, with conjugate $q = p/(p-1)$, so $(p-1)q = p$. Each term is at most $\|X\|_p\|X+Y\|_p^{p/q}$ or the same with Y. Thus $E|X+Y|^p \le (\|X\|_p + \|Y\|_p)\,(E|X+Y|^p)^{1/q}$. Divide by $(E|X+Y|^p)^{1/q}$. Since $1 - 1/q = 1/p$, the left side becomes $\|X+Y\|_p$.

Tail identities. Nonnegative integer-valued: $EX = \sum_{k\ge1} P(X \ge k)$. Nonnegative continuous: $EX = \int_0^\infty (1 - F(x))\,dx$, and $EX^k = \int_0^\infty kx^{k-1}(1-F(x))\,dx$.

2. A random sample (5.1–5.4)

A **random sample** is $X_1, \ldots, X_n$ i.i.d. with pdf or pmf $f(x \mid \theta)$. A **statistic** is any function of the data alone (no unknown θ left inside). Its distribution is the **sampling distribution**. An estimator is a statistic used as a guess for θ; the number you get from data is the estimate.

Sample mean and variance. $\bar X = n^{-1}\sum X_i$, $S^2 = (n-1)^{-1}\sum (X_i - \bar X)^2$. Always $E\bar X = \mu$ and $\mathrm{Var}(\bar X) = \sigma^2/n$. If the variance exists, $ES^2 = \sigma^2$ (the $n-1$ is what makes it unbiased). The version with n in the denominator is biased low.

Sums. The mgf of an independent sum is the product of the mgfs. Concretely: independent Poissons add their λ; independent normals add means and variances; Gamma(α_i, β) with the *same* scale β add their shapes; n i.i.d. Exp(β) (mean β) sum to Gamma(n, β).

Normal sample. If X_i i.i.d. $N(\mu, \sigma^2)$, then $\bar X \sim N(\mu, \sigma^2/n)$, $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$, and $\bar X$ is independent of S^2. Therefore

$$\frac{\sqrt{n}\,(\bar X - \mu)}{S} \sim t_{n-1}.$$

Building blocks: Z_i i.i.d. $N(0,1)$ give $\sum Z_i^2 \sim \chi^2_n$; $Z/\sqrt{V/\nu} \sim t_\nu$ if $V \sim \chi^2_\nu$ is independent of Z; $(U/\nu_1)/(V/\nu_2) \sim F_{\nu_1,\nu_2}$ for independent chi-squares.

Order statistics. Write $X_{(1)} < \cdots < X_{(n)}$. On the region $x_1 < \cdots < x_n$ the joint density is $n!\prod f(x_i)$. The kth has density

$$f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} F(x)^{k-1}(1-F(x))^{n-k} f(x).$$

For Unif$(0,1)$, $X_{(k)} \sim$ Beta$(k, n-k+1)$. The sample max has cdf F^n; the sample min has cdf $1 - (1-F)^n$.

3. Convergence (5.5)

Four modes. $X_n \to X$ **in probability** if $P(|X_n - X| > \epsilon) \to 0$ for every $\epsilon > 0$. **Almost surely** if $P(X_n \to X) = 1$. **In distribution** if $F_{X_n}(t) \to F_X(t)$ at every continuity point of F_X (equivalently $Eg(X_n) \to Eg(X)$ for every bounded continuous g). **In** L^p ($p \ge 1$) if $E|X_n - X|^p \to 0$.

What implies what. Almost sure $\Rightarrow$ in probability $\Rightarrow$ in distribution. Also $L^p \Rightarrow$ in probability, by Markov: $P(|X_n - X| \ge \epsilon) \le E|X_n - X|^p/\epsilon^p$. The reverse arrows are false.

Counters. L^p does not give almost sure: independent $X_n = 1$ with probability $1/n$ and 0 otherwise. Then $E|X_n|^p = 1/n \to 0$, but $\sum P(X_n \ne 0) = \infty$, so $X_n = 1$ infinitely often. Almost sure does not give L^p: independent $X_n = 2^n$ with probability 2^{-n}, else 0. Then $\sum 2^{-n} < \infty$, so $X_n \to 0$ a.s., but $E|X_n| = 1 \not\to 0$. (The homework display $X_n = n^2$ with probability 2^{-n} *does* go to 0 in L^p, because $n^{2p}/2^n \to 0$. The height has to be exponential.)

Tools. Continuous mapping: g continuous and $X_n \to X$ in distribution, probability, or a.s. $\Rightarrow g(X_n) \to g(X)$ in the same mode. Slutsky: $X_n \to_d X$ and $Y_n \to_p c$ give $X_n + Y_n \to_d X + c$, $X_nY_n \to_d cX$, and $X_n/Y_n \to_d X/c$ if $c \ne 0$.

LLN and CLT. Iid with finite mean: $\bar X \to_p \mu$ (WLLN); with finite mean, $\bar X \to \mu$ almost surely (SLLN). Iid with mean μ and variance $\sigma^2 \in (0,\infty)$:

$$\sqrt{n}\,(\bar X - \mu)/\sigma \ \to_d \ N(0,1).$$

Delta method. If $\sqrt{n}\,(Y_n - \theta) \to_d N(0, \sigma^2)$ and $g'(\theta) \ne 0$, then

$$\sqrt{n}\big(g(Y_n) - g(\theta)\big) \to_d N\big(0,\ \sigma^2[g'(\theta)]^2\big).$$

If $g'(\theta) = 0$ but $g''(\theta) \ne 0$, the $\sqrt{n}$ rate is too fast and the limit is a chi-square, not a normal:

$$n\big(g(Y_n) - g(\theta)\big) \to_d \tfrac{1}{2}\sigma^2 g''(\theta)\,\chi^2_1.$$

Bernoulli variance: $g(p) = p(1-p)$, $g' = 1 - 2p$, $g'' = -2$, and $\sigma^2 = p(1-p)$. For $p \ne 1/2$ the normal limit has variance $(1-2p)^2p(1-p)$. For $p = 1/2$, $g' = 0$, so $n\big(Y_n(1-Y_n) - 1/4\big) \to_d -\frac{1}{4}\chi^2_1$.

4. Drawing a sample (5.6)

Inverse transform. Draw $U \sim$ Unif$(0,1)$ and set $X = F^{-1}(U)$, where $F^{-1}(u) = \inf\{x : F(x) \ge u\}$. If F is continuous and strictly increasing, this is the ordinary inverse, and $F(X) \sim$ Unif$(0,1)$.

Left-truncated normal $N^+(\mu, c, \sigma^2)$, density proportional to a $N(\mu, \sigma^2)$ density on $[c, \infty)$. With $\delta = (c - \mu)/\sigma$,

$$F(x) = \frac{\Phi((x-\mu)/\sigma) - \Phi(\delta)}{1 - \Phi(\delta)}, \quad x \ge c,$$

$$F^{-1}(u) = \mu + \sigma\,\Phi^{-1}\big(\Phi(\delta) + u(1 - \Phi(\delta))\big).$$

Accept–reject. Target f, easy proposal g, with $f \le Mg$. Draw $Y \sim g$ and $U \sim$ Unif$(0,1)$; accept Y if $U \le f(Y)/(Mg(Y))$. The acceptance probability is $1/M$, so the best g (or best parameter inside a family) is the one with the smallest $M = \sup f/g$. For a left-truncated normal and $g(x) = \alpha e^{-\alpha(x-c)}1_{\{x \ge c\}}$, look at $\log(f/g)$. Its derivative is

$-(x-\mu)/\sigma^2+\alpha$, so the ratio peaks at $x^\star=\mu+\alpha\sigma^2$ when that point is $\geq c$, and at the boundary $x=c$ otherwise. Compute $M(\alpha)$ in each case and minimize it in α.

5. Likelihood principle (6.3)

The **likelihood** is $L(\theta \mid x)=f(x \mid \theta)$ viewed as a function of θ with the data held fixed. Two likelihoods that differ by a factor not depending on θ carry the same information about θ.

The principle. If sample x from one experiment and sample y from another satisfy $L(\theta \mid x)=c(x,y)L(\theta \mid y)$ with c free of θ, then any conclusion about θ should be the same from x and from y.

Why it is forced. The sufficiency principle says a sufficient statistic is enough. The conditionality principle says that if a coin-flip independent of θ chooses which experiment you run, you should condition on the experiment that actually happened. Birnbaum's theorem: those two principles together imply the likelihood principle.

Binomial vs negative binomial. Fixed n, observe s successes: likelihood $\propto \theta^s(1-\theta)^{n-s}$. Fixed s successes, observe the number of trials: same kernel in θ. The likelihood principle says the inference for θ is the same. A frequentist p-value can still differ, because it uses the whole sample space of the experiment that was planned, not only the observed likelihood.

6. Finding an estimator (7.1, 7.2.1–2)

A point estimator is a statistic $W(X_1,\dots,X_n)$ meant to be close to θ. "Close" is measured in Section 7.

Method of moments. Compute the first k population moments $E_\theta X^j$ ($k=\dim\theta$). Set them equal to the sample moments $n^{-1}\sum X_i^j$ and solve for θ. One parameter usually means solve $\bar X=\mu(\theta)$. The estimator is consistent when μ is nicely invertible, because $\bar X\to_p\mu(\theta)$ and the continuous-mapping theorem applies. It can fall outside the parameter space, and it ignores the rest of the likelihood.

Maximum likelihood. Maximize $L(\theta)=\prod_i f(X_i \mid \theta)$, or equivalently $\ell(\theta)=\sum\log f(X_i \mid \theta)$. Differentiate, solve $\partial\ell/\partial\theta=0$, and check it is a maximum (second derivative negative, or the log likelihood goes to $-\infty$ at the boundary). If the *support depends on* θ (Uniform$(0,\theta)$, shifted exponential), do not trust the derivative: the maximum is usually at the boundary. For Unif$(0,\theta)$ the MLE is $X_{(n)}$.

Invariance. If $\widehat\theta$ is the MLE of θ and τ is any function, the MLE of $\tau(\theta)$ is $\tau(\widehat\theta)$. If τ is not one-to-one, use the largest likelihood among θ with $\tau(\theta)=\eta$.

Pareto recipe (HW2). $f(x \mid \alpha)=\alpha x^{-(\alpha+1)}1_{\{x\geq 1\}}$. Then $\ell=n\log\alpha-(\alpha+1)\sum\log X_i$, so

$$\widehat\alpha=\frac{n}{\sum_{i=1}^n\log X_i}.$$

($\log X_i$ is exponential with mean $1/\alpha$, so $E\widehat\alpha=n\alpha/(n-1)$: the MLE is biased.)

Usual MLEs. Bernoulli or Binomial proportion: $\bar X$. Poisson λ: $\bar X$. Normal mean, σ known or not: $\bar X$. Normal variance: $\sum(X_i-\bar X)^2/n$ (divide by $n-1$ to unbias it). Exponential mean β: $\bar X$.

7. Is the estimator good? (7.3.1–2)

MSE. $\mathrm{MSE}(\theta)=E_\theta(W-\theta)^2=\mathrm{Var}_\theta(W)+(\mathrm{Bias})^2$, with Bias $=E_\theta W-\theta$. Unbiased means the bias term is 0, so MSE equals variance. To compare two estimators, plot or subtract these functions of θ. There is no estimator with smallest MSE at every θ: the constant $W\equiv\theta_0$ has MSE 0 at θ_0 and is terrible elsewhere. Bernoulli MOM and MLE are both $\bar X$, with MSE $=\theta(1-\theta)/n$.

Best unbiased. Among estimators with $EW=\tau(\theta)$ for every θ, a best one has the smallest variance at every θ.

Cramér–Rao. Assume the support does not depend on θ and you may differentiate under the integral. For unbiased W of $\tau(\theta)$,

$$\mathrm{Var}_\theta(W)\ \geq\ \frac{[\tau'(\theta)]^2}{I_n(\theta)}.$$

For $\tau(\theta)=\theta$ the numerator is 1. Fisher information of one observation, and of the whole iid sample, is

$$I_1(\theta)=E_\theta\Big(\frac{\partial}{\partial\theta}\log f(X \mid \theta)\Big)^2=-E_\theta\Big(\frac{\partial^2}{\partial\theta^2}\log f(X \mid \theta)\Big),$$

and $I_n(\theta)=nI_1(\theta)$. The two formulas agree when the regularity conditions hold; the second is usually the one you compute. Also $E(\partial\ell/\partial\theta)=0$.

When the bound is achieved. Equality holds iff the score is linear in the estimator:

$$\frac{\partial}{\partial\theta}\log L(\theta \mid x)=a(\theta)\big(W(x)-\tau(\theta)\big).$$

Then W is best unbiased for $\tau(\theta)$. Exponential families usually attain the bound for the mean of their sufficient statistic. If the support depends on θ, CRLB does not apply; do not quote it for Unif$(0,\theta)$.

Pareto information. $\partial^2\ell/\partial\alpha^2=-n/\alpha^2$, so $I_n(\alpha)=n/\alpha^2$ and any unbiased estimator of α has variance at least α^2/n. The MLE $n/\sum\log X_i$ does not attain it: it is biased, with mean $n\alpha/(n-1)$.

One-parameter informations I_1. Bernoulli(p): $1/(p(1-p))$. Poisson(λ): $1/\lambda$. $N(\mu,\sigma^2)$ for μ, σ known: $1/\sigma^2$. Exponential with mean β: $1/\beta^2$. Pareto shape α on $[1,\infty)$: $1/\alpha^2$. Multiply by n for the sample. The unbiased estimator $(n-1)/\sum\log X_i$ of the Pareto shape has variance $\alpha^2/(n-2)$ for $n>2$, which sits above α^2/n.

Scaling an estimator. If $EW=\theta$ and you use aW, then bias is $(a-1)\theta$ and $\mathrm{MSE}=a^2\mathrm{Var}(W)+(a-1)^2\theta^2$. This is the comparison when MOM and MLE differ by a factor such as $n/(n-1)$.

A question, part by part. (i) A probability bound with a sample size: Markov or Chebyshev, and check whether the bound gets worse at an endpoint such as $p=a$. (ii) A convergence arrow: quote the chain, or give one of the two counters; if a derivative appears, check whether it is zero before using the ordinary delta method. (iii) "Simulate": inverse cdf if you can invert F; accept–reject if you are handed a proposal, and optimize the proposal by minimizing $\sup f/g$. (iv) "Find an estimator": MOM is one moment equation; MLE is the log likelihood, with a boundary check. (v) "Which is better": write bias and variance, add them. (vi) "How good can an unbiased estimator be": $1/I_n(\theta)$, after checking the support does not depend on θ.